Weekly update 7

DRIFT: Detecting and Retiring Invalidated Facts over Time

Week 6 feedback addressed: new name, pipeline, correctness of the forgetting policies, checkpoint 2 across all 1,000 weightings

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Previously (Week 6), and the feedback this week addresses

WEEK 6 final run complete: 7 policies x 5 answer models x 3 benchmarks, 105 cells, 10,360 graded answers; all four RQs answered.

- **P2 best on both fact benchmarks:** accuracy 93.5% → 98.5% and 67.4% → 79.4%.
- **Small models gain most;** the timeline, not the graph, is what helps.
- **RQ4:** rankings transfer to multi-agent memory.
- **ConvoMem:** negative result, limited by extraction.

feedback from the meeting	slide
Name and story: does DRIFT fit the thesis?	2
Pipeline and workflow, general takeaways	3
What exactly is a forgetting policy?	4
RQ2 as a bar chart, P2 highlighted	5
RQ3 in percentages, y-axis to 100%	6
Checkpoint 3: 'close to' the frontier	7
Spearman for RQ4: formula, why, reference	8
Checkpoint 2: all 1,000 weightings, mean, variance, min, max	9, 10
Justify correctness: empirical and theoretical	11, 12
Observe decay; ConvoMem retrieval question	13, 14

All values are the final run. Accuracy and staleness are shown as percentages throughout.

A name that tells the thesis story

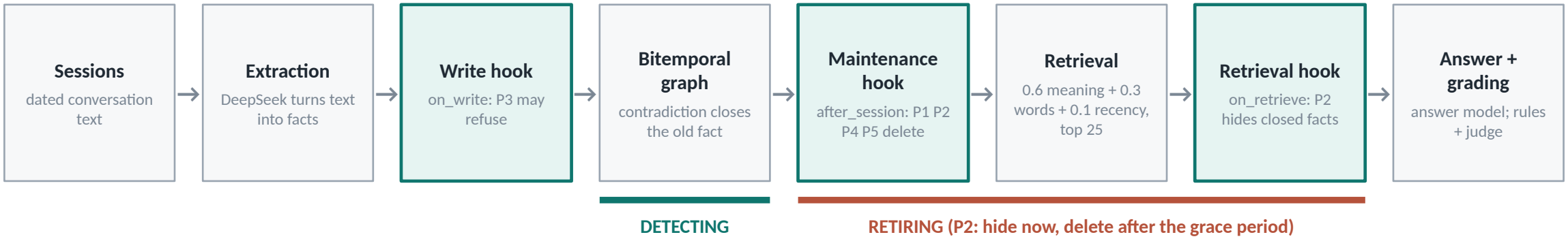
WHY CHANGE 'Decay Rates' suggests memory should fade with age, but deleting by age (P1) made memory worse. What helps is removing facts that are no longer true.

word	what it is in DRIFT	where it is measured
Detecting	noticing that a new fact replaces a stored one (same subject and relation, new value)	matching study: rules catch 95 of 96 changes; LLM check needed across writers
Retiring	hiding the replaced fact from retrieval at once, deleting it after the grace period	P2 vs the other five policies; grace-period study
Invalidated Facts	facts whose validity window has closed: still stored, no longer current	staleness: answers that repeat an invalidated value
over Time	the bitemporal timeline: when a fact was true, and when the agent learned it	P2 vs P5 (same graph, no dates): +23.5 and +15.2 points

The story in one line: memory should detect when a fact has been replaced and retire the old one using its timeline; the thesis measures how much that matters, compared with five other forgetting rules.

The DRIFT pipeline: everything is fixed except the forgetting policy

HOOKS the three highlighted points are where a forgetting policy can act; every other stage is identical in every experiment.



One question through the pipeline (MQuAKE-T, Llama 3.1 8B)

	what reaches the prompt	answer	prompt tokens
P0 append-only	Hipkins (29 Jan) and Ardern (7 Jan); 8 of 25 lines outdated	Ardern (stale)	609
P2 bitemporal	Hipkins only; 0 outdated lines	**Hipkins (correct)**	531

General takeaways: retrieval cannot separate old from new (they differ in one value); the policy must. Hiding replaced facts removes the problem before the model sees it, and also cuts prompt tokens.

What a forgetting policy is: refuse, hide or delete (closing is not forgetting)

DEFINITION a forgetting policy decides which stored facts the answer model can no longer see. Closing a fact's validity window happens in the store for every policy, including P0, which forgets nothing.

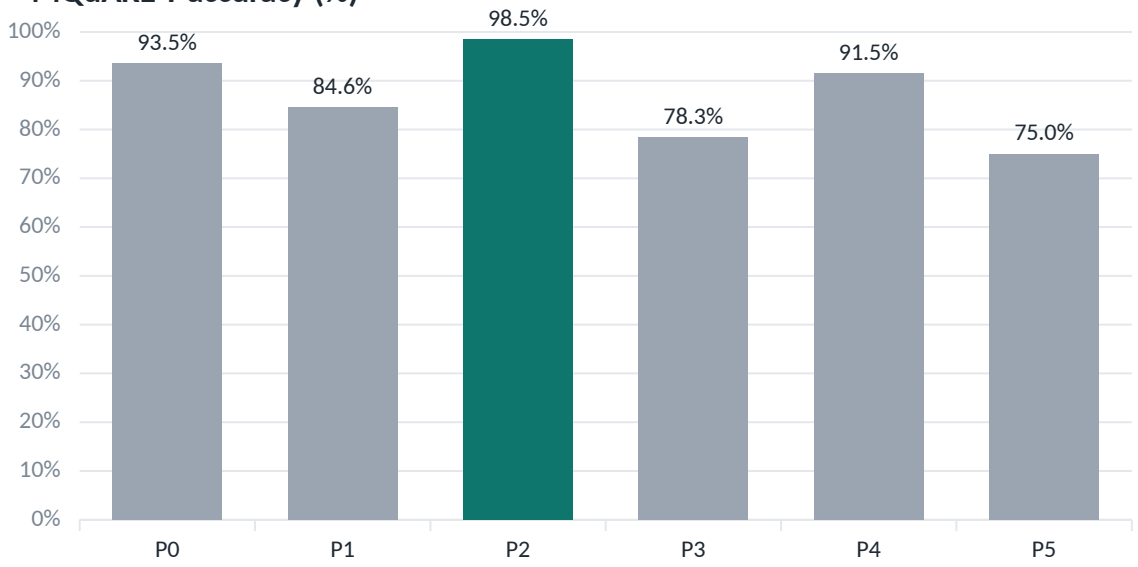
action	when	reversible?	policies	example
close (not forgetting)	a later fact replaces it	yes: still stored and visible	all, in the store	Ardern fact gets an end date
refuse	at write	no: never stored	P3	the Hipkins correction is rejected as a duplicate
hide	at retrieval	yes	**P2 (closed facts), NOMEM (all)**	Ardern stays stored, never shown
delete	after a session	no	P1 (age), P2 (after grace), P4 (LLM), P5 (support)	Ardern removed after 7 days (P2)

Why P2 hides first and deletes later: hiding gives the accuracy benefit immediately and is reversible, so questions about the past can still be answered; deleting saves storage once the grace period has passed (slide 13).

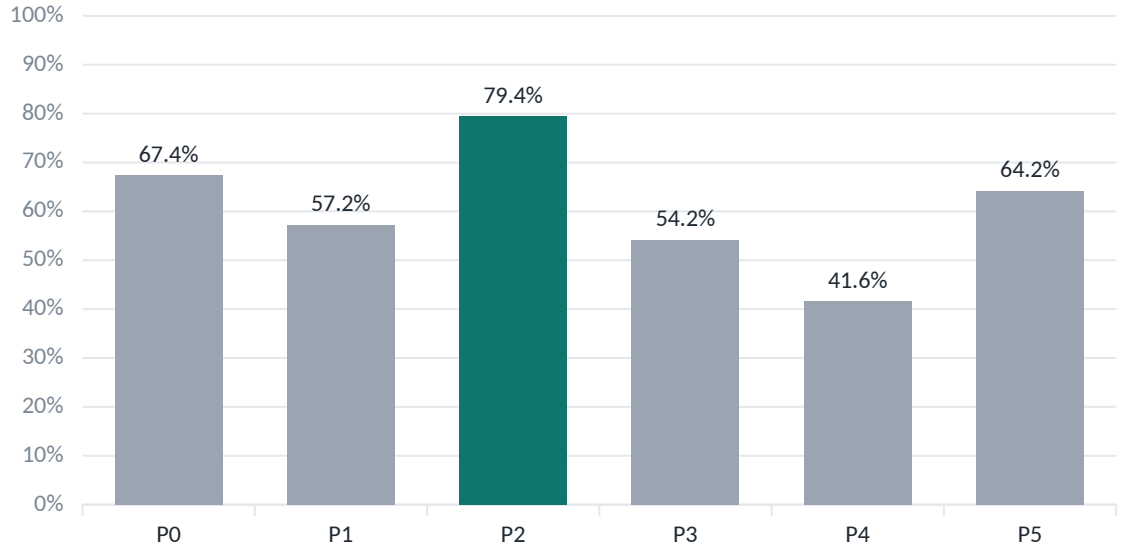
RQ2: P2 is the most accurate policy on both fact benchmarks

HOW TO READ accuracy by policy, mean over the five answer models; P2 highlighted. Gains are in percentage points; every one has p < 0.01 after Holm correction.

MQuAKE-T accuracy (%)



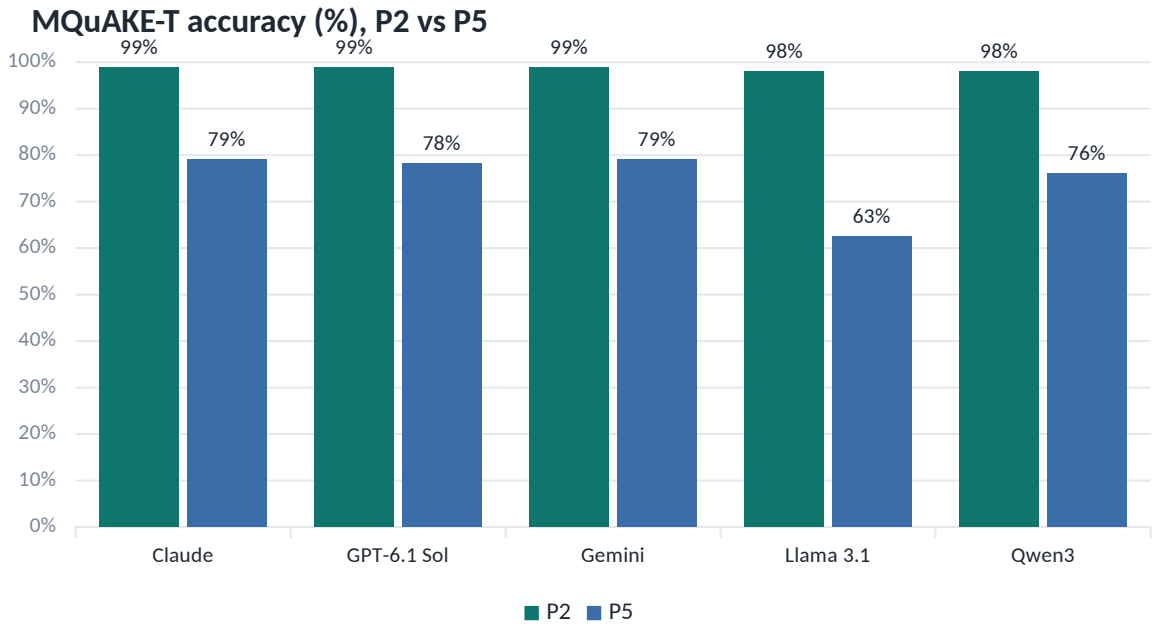
FactConsolidation accuracy (%)

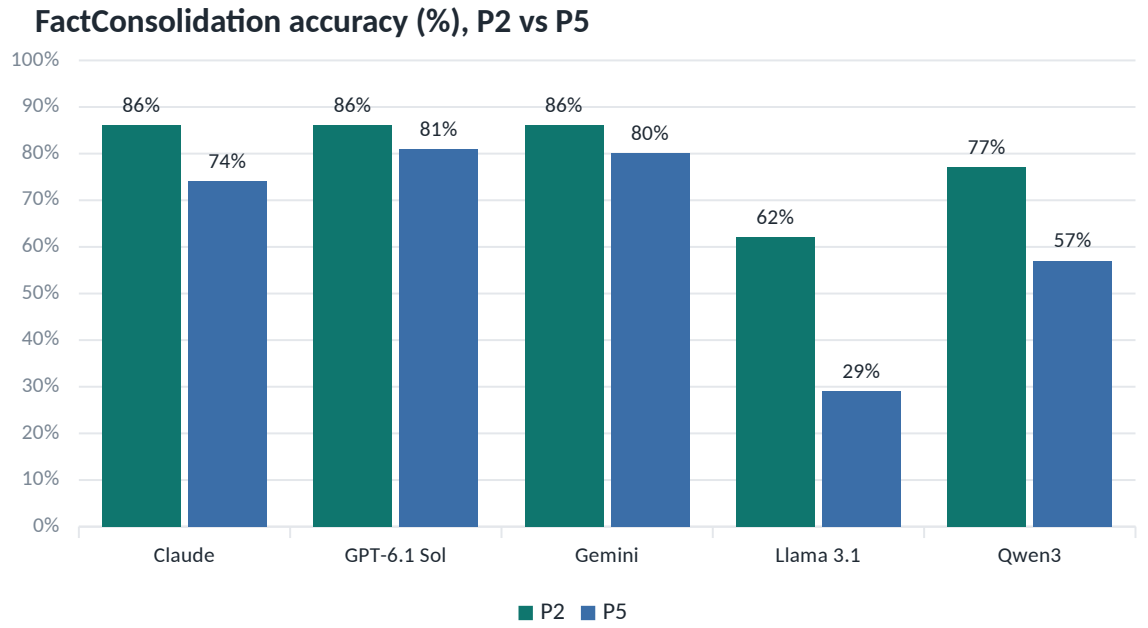


P2 gain over	P0 keep all	P1 time decay	P3 novelty gate	P4 LLM review	P5 topology
MQuAKE-T	+5.0 pts	+14.0 pts	+20.2 pts	+7.1 pts	+23.5 pts
FactConsolidation	+12.0 pts	+22.2 pts	+25.2 pts	+37.8 pts	+15.2 pts

RQ3: it is the timeline that helps, not the graph

P5 sees the same facts and graph as P2 but not the dates, so it must guess which of two conflicting facts is current.



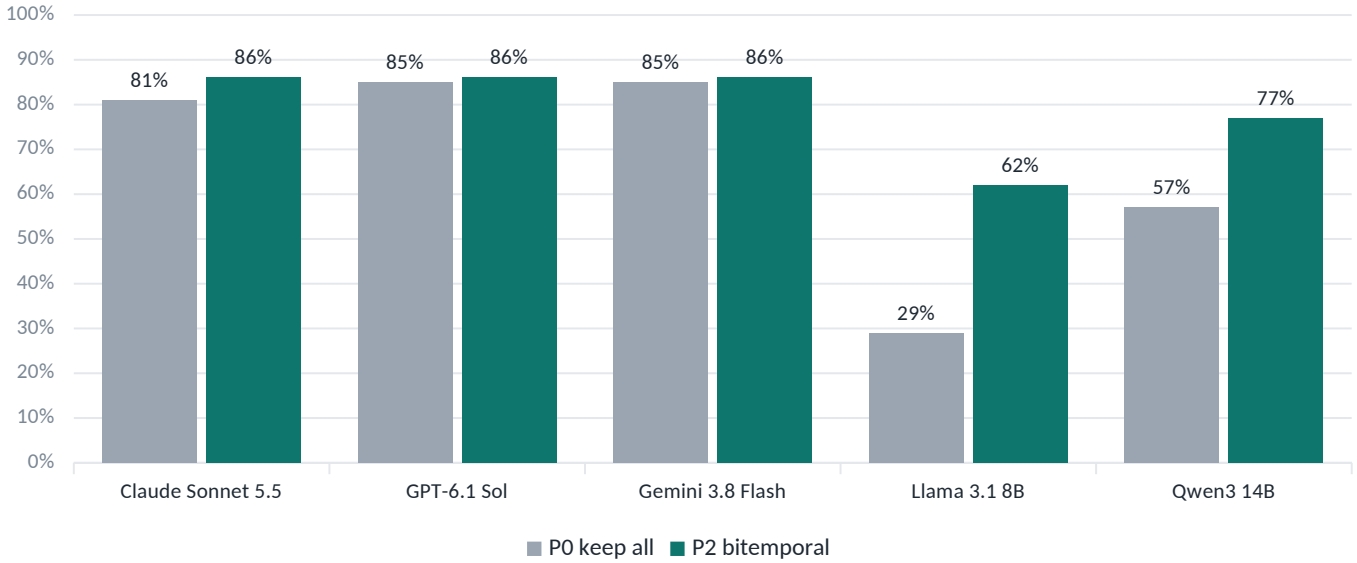


Answer: P2 beats P5 for every model: +23.5 points on MQuAKE-T (113 questions won, 0 lost) and +15.2 points on FactConsolidation, both $p < 0.001$. Structural support favours the value mentioned longest, which is the old one.

Checkpoint 3: with P2, small models come close to the frontier

COST PER CORRECT ANSWER list-price USD for 1,000 correct answers under P2 on MQuAKE-T.

FactConsolidation accuracy (%), P0 → P2



model	P2 acc. MQuAKE-T	$ / 1k correct
Claude Sonnet 5.5	99.0%	$1.73
GPT-6.1 Sol	99.0%	$1.10
Gemini 3.8 Flash	99.0%	$0.47
Qwen3 14B	97.9%	$0.12
Llama 3.1 8B	**97.9%**	**$0.01**

Answer: P2 adds +12.0 points (MQuAKE-T) and +26.5 points (FactConsolidation) for small models, against +0.5 and +3.0 for frontier models. Llama 3.1 8B comes close to the frontier on real updates (97.9% vs 99.0%) at about 1/160 of Claude's cost per correct answer; a gap remains on invented facts (62% vs 86%).

RQ4: Spearman correlation, what it measures and why it is used

SPEARMAN'S ρ compares two rankings of the same items: 1 = identical order, 0 = unrelated, −1 = reversed (Spearman, 1904, American Journal of Psychology 15(1), 72 to 101).

- **Formula:** $\rho = 1 - 6 \cdot \Sigma d^2 / (n(n^2 - 1))$, where d is a policy's rank difference and n = 6 policies.
- **Why this metric:** RQ4 asks whether the order of policies holds, not whether the scores stay the same; Spearman uses ranks only, so a lower overall accuracy does not count as disagreement.
- **What it measures:** agreement between the single-agent ranking and each multi-agent ranking (accuracy).

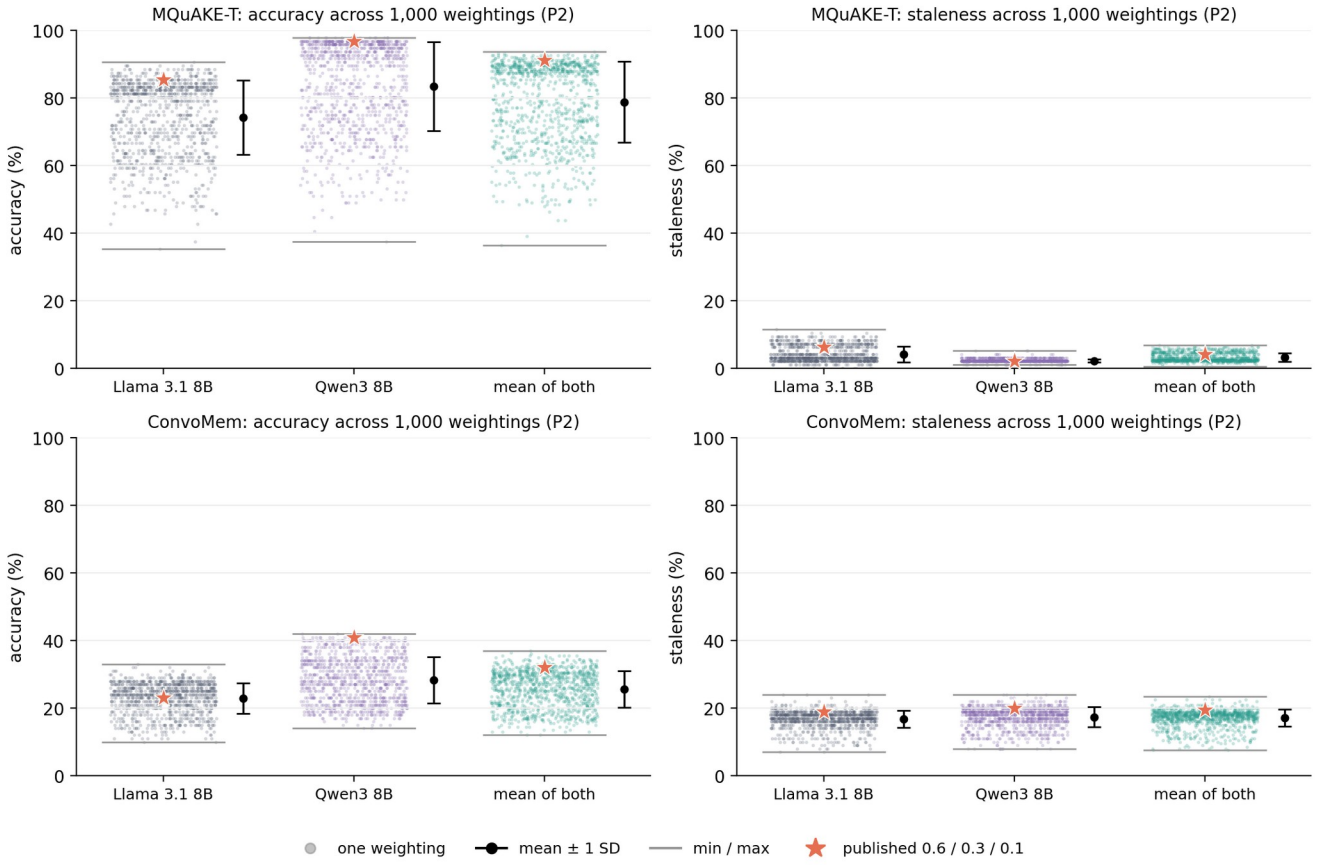
rank	single agent	independent agents
1	P2 (98.5%)	P2 (73.3%)
2	P0 (93.5%)	P0 (73.1%)
3	P4 (91.5%)	P4 (72.3%)
4	P1 (84.6%)	P1 (64.8%)
5	P3 (78.3%)	**P5 (64.0%)**
6	P5 (75.0%)	**P3 (63.5%)**

Worked example: only P3 and P5 swap, so $\Sigma d^2 = 1 + 1 = 2$ and $\rho = 1 - 12/210 = 0.94$.

Result: ρ = 0.94 (independent), 0.93 (orchestrator, shared): the policy order transfers; P2 is best or tied-best everywhere. With only 6 items, one swapped pair costs about 0.06.

Checkpoint 2: results across all 1,000 retrieval weightings

THE SWEEP every weighting with semantic, lexical and recency each from 0.1 to 1.0 (10 x 10 x 10), P2, two small models, MQuAKE-T and ConvoMem: 37,572 graded answers.



mean of both models	mean	SD	min	max	published
MQuAKE-T acc.	78.8%	12.0%	36.5%	93.8%	**91.1%**
ConvoMem acc.	25.6%	5.4%	12.0%	37.0%	**32.0%**
MQuAKE-T stale	3.2%	1.2%	0.5%	6.8%	4.2%
ConvoMem stale	17.1%	2.5%	7.5%	23.5%	19.5%

- **Accuracy:** only 6.5% (MQuAKE-T) and 7.9% (ConvoMem) of weightings score higher; none significantly after Holm.
- **Variance:** 0.0144 and 0.0029 (accuracy as a fraction).
- **Staleness** is higher at 0.6/0.3/0.1 than with heavy recency, which costs accuracy; P2, not the weights, removes staleness.

Checkpoint 2: the retrieval weights hold on unseen data

RETRIEVAL SCORE 0.6 x cosine (meaning) + 0.3 x word overlap + 0.1 x recency; the 25 best-scoring facts reach the prompt.

weighting (FactConsolidation, P2)	Llama 3.1 8B	Qwen3 14B
0.6 / 0.3 / 0.1 (thesis setting)	62%	77%
0.9 / 0.1 / 0.2 (MQuAKE-T winner)	62% (p = 1.0)	81% (p = 0.34)
0.8 / 0.1 / 0.1 (ConvoMem winner)	56% (p = 0.21)	78% (p = 1.0)

- **Search (Week 5):** 1,000 weightings, none better after Holm correction.
- **Held-out (now):** FactConsolidation was never searched; neither winner is significantly better.
- **Holm correction:** adjusts p-values when many weightings are tested, so luck does not look like a win.

What I will claim: not 'the best weights' but a robust setting: not beaten anywhere it has been tested, inside the safe region (recency share at most 0.2), and confirmed on data it was never tuned on.

Is a forgetting policy correct? Definition and theoretical analysis

CORRECT sound = never hides a fact that is still true (protects accuracy); complete = hides every fact that is no longer true (protects against staleness).

policy	what the rule decides on	prediction from the rule
P2 bitemporal	has a later fact replaced this one?	sound and complete whenever contradiction detection is correct; its errors are detection errors
P1 time decay	is the fact older than 20 days?	age is not truth: deletes true old facts, keeps recent replaced ones
P3 novelty gate	is the new fact too similar to a stored one?	a correction above 0.9 similarity can never be stored
P4 LLM consolidation	does the model believe the fact?	sound only when the world matches the model's training
P5 topology	which value has more support in the graph?	no dates, so it favours the longest-mentioned (old) value

Staleness bound: P(stale) = P(old value in the prompt) x P(model repeats it). The sweep measured P(stale | old value not shown) = 0%, so a complete policy gives zero staleness for any model; that is why small models gain most.

Correctness, empirically: P2 never hid a true fact

MEASURED for every changed fact, the memory at the end of the run: is the old value still visible, is the current value still visible? Mean over the five models.

policy	MQuAKE-T complete	MQuAKE-T sound	FactCons. complete	FactCons. sound
P0 append-only	0.0%	100.0%	0.0%	100.0%
P1 time decay	41.7%	89.6%	61.5%	82.8%
P2 bitemporal	**96.9%**	**100.0%**	**82.7%**	**100.0%**
P3 novelty gate	1.0%	87.5%	1.9%	75.0%
P4 LLM consolidation	0.0%	99.0%	0.4%	47.2%
P5 topology	1.9%	81.0%	3.8%	92.2%

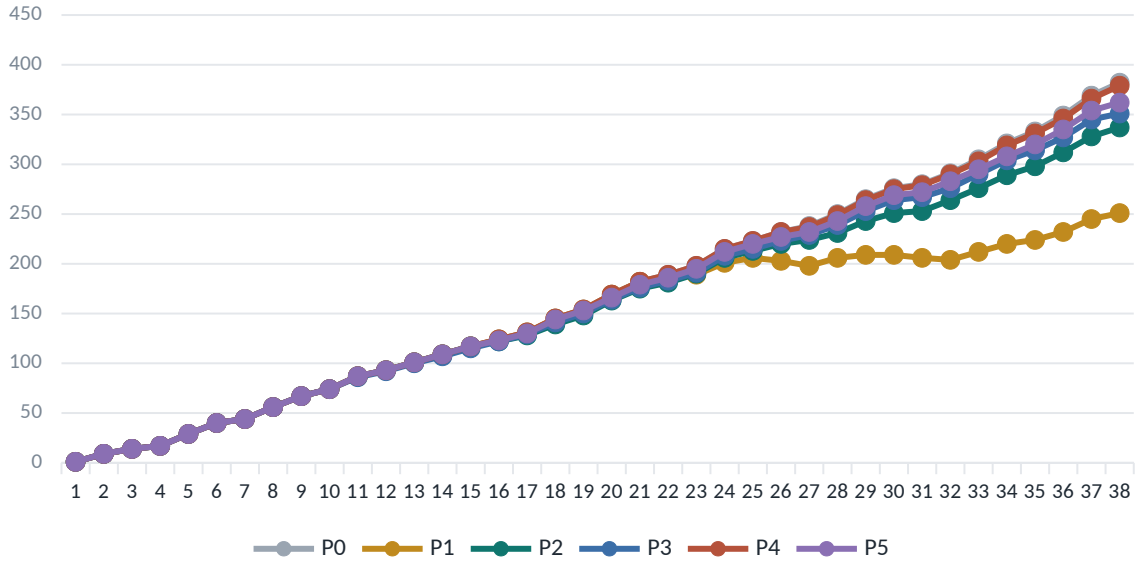
- **P2:** 100% sound on both; 96.9% and 82.7% complete. The misses are relation names written two ways (created_in_country vs created_in), which the LLM check fixes.
- **P3** loses corrections (75 to 88% sound); **P4** deleted half the true values on FactConsolidation (47.2% sound).
- **P5 and P4** almost never retire the old value (0 to 4% complete).

Exception, ConvoMem: false contradictions make P2 unsound in dialogue: it hid 4 of the 14 relevant current facts, which matches its 3.8-point accuracy loss. Values matched by rule from the saved graphs; treat as close estimates.

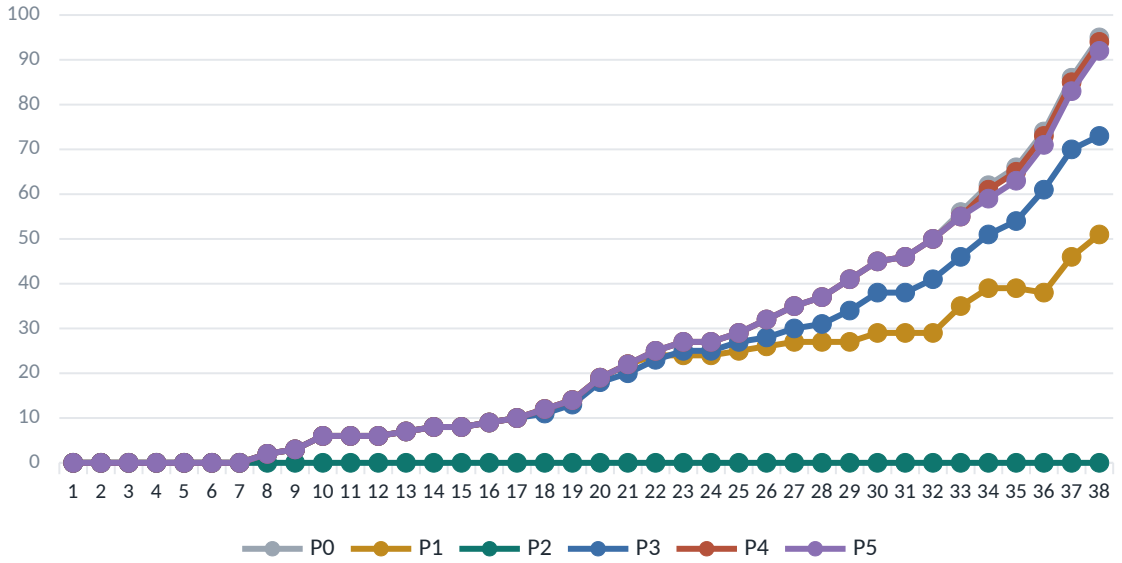
Observing decay: how each policy's memory evolves over 38 sessions

OUTDATED AND VISIBLE facts whose validity has closed but which retrieval can still return. P2 hides them, so the count it exposes is zero by design (MQuAKE-T).

Facts kept in memory



Outdated facts still visible to retrieval



Grace period (P2): current accuracy stays 96.9% to 97.9% at every setting (Qwen3 14B); history accuracy rises 12.5% (0 days) → 53.1% (7) → 90.6% (21) → 99.0% (never), with storage 296 → 382 facts.

ConvoMem: is retrieval the reason it fails?

CHECKED where each ConvoMem question is lost, from the saved final-run graphs and the 1,000-weighting sweep.

stage (100 questions, mean over models)	P0	P2
current value anywhere in memory	30%	30%
… in a fact about the question	**14%**	14%
… reaches the top 25 retrieved	14%	**10%**
answered correctly	11.0%	7.2%

- **Extraction is the ceiling:** the answer is stored for only 14 of 100 questions.
- **Retrieval ranking loses nothing:** under P0 all 14 reach the prompt; the best of 1,000 weightings finds the gold value for 58% vs 57% for 0.6/0.3/0.1 (sweep).
- **P2's loss happens at retrieval time:** its filter hid 4 of the 14 after false contradictions.

Answer: retrieval ranking is not the bottleneck; extraction is. P2's 3.8-point loss is its retrieval filter acting on false contradictions, so the fix is better detection and extraction in dialogue, not different weights.

Thesis status and next steps

THESIS the final document is being written by me, line by line, in Overleaf on the Stevens template.

item	status
Overleaf project: chapters, tables and final-run figures in place	**done**
Title updated to Detecting and Retiring Invalidated Facts over Time	**done**
Chapters 1 to 3 rewritten in my own words with the comments addressed	in progress
Chapter 4 results and Chapter 5 analysis, including correctness and the 1,000-weighting sweep	in progress
Chapter 6 discussion and conclusion	next
Defense: 3 December, 2:00 pm (Zoom)	**date set**

> **Questions for the committee:** is the new title acceptable, and is the soundness and completeness analysis the right form of correctness argument for Chapter 5?